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Eye of Horus



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I. Present and Justify a Problem and Solution Requirements

Pollution

Pollution is the increase of harmful environmental materials, including the soil, air, water, [Fig (1.1)], and many more. These materials can be natural or man-made, and

they are called pollutants. Natural pollution, such as volcanic ash, can't be controlled. Meanwhile, human activities leading to pollution, such as factory runoff, can be controlled. Still, most countries don't take solving the pollution problem

into consideration. In the majority, all human activity produces pollution as car fumes, burning coal for electricity, factories and homes, garbage and sewage that pollutes the waters all around the world.

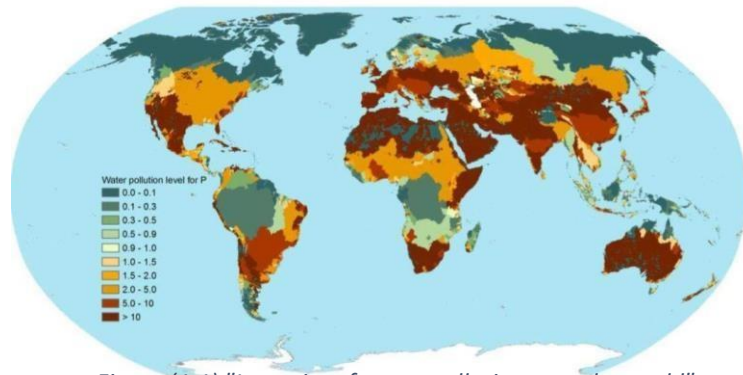


Figure (1.1) "Intensity of water pollution over the world"

Types of Pollution: -

Pollution is classified according to the sector of the environment that is affected.

Water pollution:

Water pollution mostly happens due to excess chemical and biological substances. These substances change the water structure.

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Usually leading to harming living organisms. Polluted water shouldn't be used for drinking, agriculture or personal uses. The effect of the polluted water depends on the concentration and the type of pollutant.

Types of water pollution: -

- **Biological water pollution** is the presence of microorganisms that cause diseases and harm living organisms, plants, animals and humans. The source of these pollutants is infected animals and humans that transmit the infection to the water and the decaying of human and animal wastes in the water. The pollutants can reach the surface and groundwater.

- **Chemical water pollution:** [Fig (1.2)] is due to the chemical substances in the water, like plant fertilization substances and pesticides. Or, it can be due to human and animal wastes.



Figure (1.2) "Water Pollution"

Air pollution:

Air pollution is the presence of high amounts of chemicals in the air [Fig (1.3)]. These chemicals cause harm to all living organisms on an extensive scale. The wind can transfer polluted air to other areas, causing global air pollution. Examples of chemical air pollutants are carbon oxides and nitrogen oxides, Sulphur oxides, Ozone and many

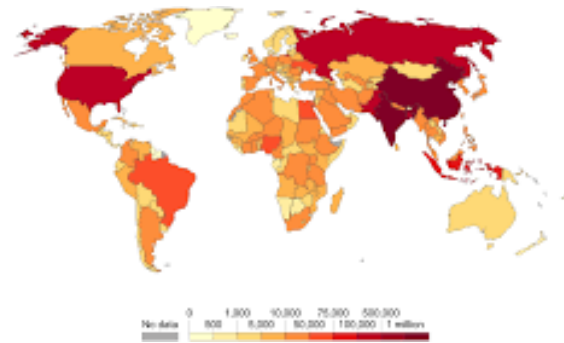


Figure (1.3) "Air pollution across the globe"

More. These gases are usually found in fresh air but become pollutants when their quantity becomes abnormal. Air pollutants may be natural, like volcanic smoke, or they can be man-made, like factory smoke, car fumes and deadly chemical gases from nuclear reactors. Some air pollutants can be seen like dark clouds of smoke. In contrast, other Air pollutants can be invisible and odorless and cause health problems without being detected.

Soil pollution:

Soil pollution is produced chiefly by houses and factories from food, garbage and other household items. When the waste decomposes, it produces a liquid called leachate, a highly toxic pollutant that causes diseases and health problems. Land pollution can be caused by water pollution, where polluted water leaks into the soil and decay, causing the inner parts of the soil to contain



Figure (1.4) "Soil pollution (trash and solid wastes)"

Harmful materials. Another form of soil pollution is trash and solid waste [Fig (1.4)]. Where junk is spread all around the world. It makes it difficult for plants to grow. And kill lots of animals.

Pollution in Egypt

Pollution has reached a far stage in Egypt. It's estimated that 2 in every 1000 die due to air pollution. That's about 35,000 a year. Egypt's air is polluted, the river is full of garbage and dead animals, and the seas are full of plastic. Due to overpopulation, there are a lot of humans that produce waste. Urban congestion leads to the concentration of pollution in areas where citizens are congested.

Causes

Natural phenomena: Natural cases are usually natural catastrophes like volcanoes, hurricanes, wind and earthquakes. These phenomena lead to water, air and soil pollution, and humans can't do anything to stop them. Natural phenomena don't cause much environmental pollution; the environment usually recovers independently.

Human activities: They are the main reason for the increase in pollution. In Egypt, Pollution mostly comes from industry, domestic sources, agriculture and transport.

Types of human activities causing pollution: -

Industry:

Most industrial processes produce pollutants, such as smoke, from their chimneys, which causes air pollution. As shown in [Fig (1.5)], the pipes transfer the factory wastes to rivers or seas, causing water pollution. Food processing, textiles, and tanneries are the factories that produce the worst pollutants .



Figure (1.5) "Pollution caused by human activities"

factories, these factories produce liquid

Pollutants into the water. These pollutants don't have treatment.

Domestic causes:

Domestic pollution comes from houses, markets and shops because of food wastes, sewage, smoke from domestic fires, and packaging materials. These pollutants are often thrown into water, burned, or buried in arid areas where all these solutions lead to pollution. To decrease this pollution, the pollutants should be recycled instead, and we won't just get rid of them. We will also benefit from the recycled materials.

Transport:

Vehicles use different kinds of petroleum. Each kind produces toxic air pollutants when burnt, and as the size of the vehicles increases, the fumes coming out increase as well [Fig (1.6)]. These toxic materials led to severe body damage, illness and sometimes death. The solution to this cause of pollution would be using electric-operated vehicles. These vehicles are already being manufactured but on a microscopic scale.



Figure (1.6) "Air pollution caused by transportation forms"

Agriculture:

Agriculture pollution happens due to plant wastes and polluted water after watering the plants and using fertilizers and pesticides. Farmers in Ethiopia use about 650,000 metric tons of fertilizers. These fertilizers will cause excess water plant growth if they reach seas or oceans. They also use over 3000 tons of different kinds of pesticides. For example, DDT is a pesticide that was once used to kill mosquitoes that carry malaria disease, which kills 1 million humans each year. Later, it was discovered that DDT harms humans and animals, especially birds. It can cause diseases and cancer, destroying the birds' eggs. DDT was banned from many countries, but it is still used to fight malaria in others.

Consequences

Diseases are the most common consequence of pollution. Diseases happen to both humans and animals. Plenty of diseases spread due to pollution. The most dangerous is cancer. Extreme and long-lasting pollution conditions lead to death. About 7 million die per year worldwide due to severe pollution conditions. The extinction of some animals because of eating garbage mistakenly or living in



Figure (1.7) "Pollution's effect on animals"

Polluted areas, [Fig (1.7)]. Devastating problems such as ozone holes have occurred due to the increased use of some chemicals as insecticides. Global warming happens due to the greenhouse gases transmitted from factories.

Fighting pollution with the help of AI

The Chinese capital, Beijing, is known for being one of the most polluted cities in the world. Although air quality has improved in the last few years, air pollution in Beijing poses a real threat to humans living there as it is surrounded by factories fueled by coal that emit harmful particulates. In 1998, air pollution was dominated by coal combustion and motor vehicles. Over the next 15 years, Beijing implemented a series of studies.



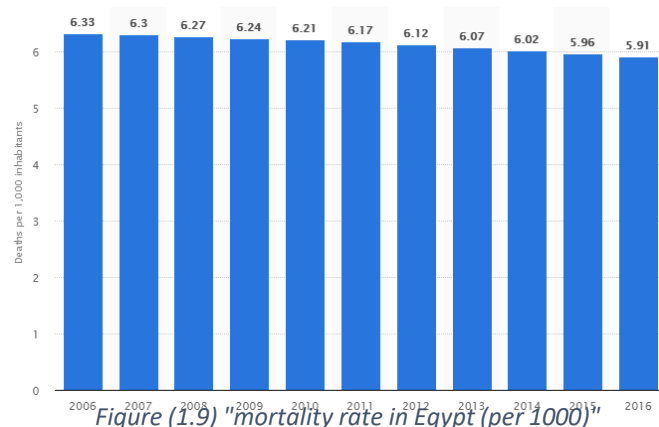
Figure (1.8) "An air quality node"

AI-based solutions were proposed, focusing on how air could be improved. The IBM research team has constructed a prototype capable of predicting air pollution using the data supplied by the Beijing Environmental Protection Bureau. Furthermore, the Egyptian Kareem Tarif and his team have invented a network of air quality nodes around Cairo [Fig (1.8)]. This network would help identify risk conditions for asthmatics in the environment.

Public health

preventing disease, improving health, prolonging life and promoting human health can be applied by analyzing the health of a population and the diseases that threaten society's education, policy-making, and research for disease and injury prevention. Public health can mobilize and engage local, state, national, and international resources to ensure the conditions in which people can be healthy. Public health presents a challenge for many countries. As a society, we should try to reach the best health conditions in Egypt and reduce its causes to ensure the conditions. Although public health standards are supposed to be applied everywhere, many areas worldwide suffer from medical deterioration.

About 600,000 persons die every year in Egypt, as shown in [Fig (1.9)], and under-five mortality is 61 deaths per 1000 children. Most of these deaths are because of medical deterioration. As a result, Egypt was delayed to the 71st place at the health level, according to the World Health Organization's statistics in 2011.



Many countries also suffer from the

Same problem, especially in developing countries. Liberia is a developing country that has significant health problems. This country suffers from epidemics and poor health care. Liberia's Ministry of Health has taken steps to solve health problems in the country. Still, diseases and the deterioration of health remain complicated problems there. On the other hand, some countries have no crucial health problems like England and France, which developing countries must learn from their experience.

Causes

Many reasons can explain why Egypt suffers from the deterioration of public health, and they are:

- **Lack of care about public hospitals:** the state doesn't care about public hospitals in terms of cleanliness, employing specialists in work and controlling people who work.

As a result of that, many mistakes happen in hospitals, and many people die because of them.

- **Lack of medicines:** A statistic in 2018 confirmed that the lack of medicines, up to 52% in the urban areas and 82% in the countryside, is a big problem, and the state must provide medicines for people.

- **Low expenditure:** Lack of health expenditure is one of the most significant health issues in Egypt as the Egyptian constitution only sets 4.9% of the GDP (gross domestic product) for health, which will be a significant impediment to developing the health system in Egypt.

- **Pollution:** Pollution badly affects public health. Contaminated water is spread throughout Egypt. This causes a lot of problems and destroys people's health. Also, contaminated air impacts public health. If the air people breathe is polluted, they will catch a lot of diseases like pulmonary diseases.

Consequences

- **Increased mortality rates:** As a result of a lack of medicines, people will not find their cure here in Egypt. They will have to travel abroad to get their treatment, but other people who are poor and can't travel will simply die. Furthermore, the death rate in hospitals increased to 1% after it was 0.84%.

- **The spread of disease:** Because of the deterioration of public health and polluted food and water, many diseases have spread in Egypt and hit people.

Artificial intelligence against public health deterioration

After the technological revolution in the 19th century, it was expected that medicine and other related fields would get the most out of the latest technologies continually discovered. Many operations and medical systems have become much easier to deal with, and many untreatable diseases in the past have become as treatable as the flu. This evolution would not be done without exploiting the capabilities of artificial intelligence. For example, physical robots are installed yearly in more than one industry to help lift, reposition and deliver supplies in factories, warehouses and hospitals. Furthermore, many projects were invented to diagnose and treat specific diseases, such as bacterial infections.

Improving the industrial base

In the 1920s, Egypt began the industrialization process after it was known for its agricultural economy since the pharaoh's era; however, Egypt is still struggling and lagging far behind the other countries that began the industrialization process much later than Egypt. The problem of the industrial sector in Egypt mainly arises from the competitive pressure facing Egyptian trademark from the main three markets in the world: countries within the MENA group, which opened up their economy and took a significant step in the road of industrialization process; countries in MENA group include Tunisia and Morocco. Second, east Asian and European transition economies are characterized by their efficient products in the international market due to the skilled labor and capital-intensive activities. Third, the world's primary prominent markets nowadays are China, India, and Bangladesh. This market is known for its sizeable unskilled labor, abundant labor, and low-wage economies.

Industrialization:

Industrialization is the process whereby a social or economic transformation alters a group or society from an agriculturally dependent society into an industrial one. In the 1920s, Egypt's economy was mainly considered agricultural since about three-quarters of the exports were raw Cotton. Similarly, Cotton was the main export.

for about more than one decade, especially in the

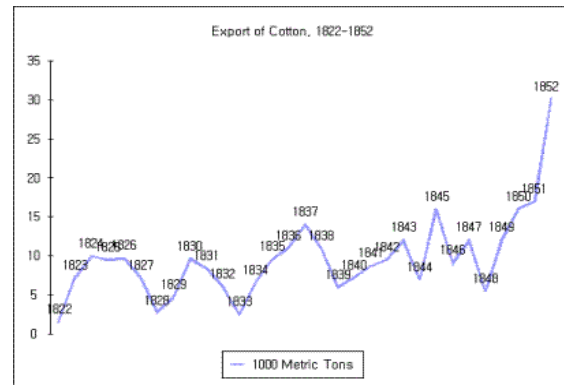


Fig (1.10) "Egypt's cotton exports"

Nineteenth century, as depicted in [Fig (1.10)]. Hence, during this period in Egypt, the industrial output focused mainly on cotton spinning and weaving, with other industries such as cigarettes and preserved food. Furthermore, the labor engaged in the industrial sector during this decade was approximately half a million out of the population capacity of 13 million, which is a relatively small number. Therefore, the Egyptian government has reached a long-term scheme to work on improving industry in Egypt.

Development of industry and sustainable development:

Before deepening the industrial base's development, it's vital to describe the concept of sustainable development. Sustainable development is defined by the World Commission on Environment and Development, WCED, as the development that meets the present without compromising the ability of the next generations to meet their own needs.

(A) Historical background of industrialization in Egypt (1800-1950):

Although Egypt was prominently known to be an agricultural country, meaning that most of the citizens' primary source of income was agriculture, at the beginning of the twentieth century, with Cotton as its main export, the initial spark in the industrialization process in Egypt had already begun in the previous decade. Egypt first knew organized manufacturing during the nineteenth century when Mohammed Ali took over. Mohammed Ali, inspired by industrial European development, sent students to learn modern applications in science. After that, Egypt started its industrial revolution by building big industries, relying on foreign expertise and technology, aside from the expert students Mohamed Ali sent to Europe. 1952, the Kingdom ended, and the Republic of Egypt was announced. In the fifties and sixties, Egypt began entrepreneurship with the Soviet Union to start up a new industrial revolution, relying mainly on governmental efforts and even the big industries that had been nationalized.

(B) Performance of the manufacturing industry in Egypt

The Egyptian manufacturing sector is concentrated mainly on seven primary industries, accounting for about 80% of establishments in the manufacturing sector. The three primary industries include textiles, food and beverages, as shown in [Fig (1.11)], and furniture, followed by non-metallic minerals, metal production, chemicals, and elemental metals. The Egyptian government has considered the The manufacturing industry is the keystone for improving the performance and quality of the industrial sector. Therefore, in 2000-2001,



Fig (1.11) "Food and beverages manufacturing process"

Manufacturing output represented approximately 20% of the gross domestic product (GDP), employing about 20% of the active manpower.

(C) Importance of Manufacturing

Manufacturing has a tremendous positive impact on the economic societies, generally, and Egypt, specifically, due to the following reasons:

- It has a significant role in decreasing the negative impacts of unemployment in Egypt and creating new jobs with appreciated wages for youth.
- Contributed to create a highly skilled and knowledgeable workforce in different industries.
- Contributed to decreasing or merely diminishing the need for imports gap.
- Increasing the production and service capacity in society.

Causes of the low-efficient industry in Egypt

Energy and foreign currency shortages are the main obstacles for Egypt's industry during this decade. Similarly, energy shortages mainly cut factory production, diminishing Egypt's exports. In contrast, currency shortages have cut the industry's ability to import raw material inputs, leading to the devaluation of the Egyptian pound in 2016.

The contribution of artificial intelligence to several industries:

It was evident that the whole world depends on new technologies in all aspects of life, and, as expected, the industrial bases all over the world are the most Beneficial. Many Technologies, especially AI, were introduced to simplify the whole process. Not only does it contribute to industries related to manufacturing and producing, but it also contributes to every known field.

Automotive Industry:

The automotive industry is one of the essentials for every single human. Everyone on earth has to ride a vehicle at least several times in his lifetime. With the aid of AI, vehicles can now save time, limit energy consumption, and dramatically reduce the more than 1.25 million deaths attributed to road traffic accidents each year. This could be applied through AI-powered software, enabling any vehicle to understand its immediate environment and navigate it safely.

Bioscience:

AI is taking bioscience to the next level by empowering researchers to conduct experiments more efficiently. With these tools at their disposal, scientists have a better chance of making breakthrough discoveries that can expand life expectancies, reduce human suffering, and create a better quality of life globally. Some companies have started to depend on AI to predict which chemical compound would achieve specific goals, such as curing a symptom or a disease.

Gaming:

Nowadays, the video game industry is no less than any other industry field. Therefore, much effort is being made to develop this field daily. Such development can be achieved by building software that empowers users to create highly complex simulations. Regarding video games, this helps developers build massive online worlds for players to explore. Perhaps more importantly, the software can also simulate scientific experiments and real-world policy questions.

Disability of Blindness

Blindness is a state of losing vision in the two eyes; a completely visually impaired individual can't see by any means. The word blindness is ordinarily utilized as an overall term for low vision or twofold understanding, which implies that even with glasses, contact focal points, medication, or medical procedure, no individual sees well; vision hindrance can go from gentle to extreme.

Visual impairment and blindness are carefully characterized as being blind in the two eyes. In any case, the word visual impairment is generally utilized as a broad term to connote visual disability or low vision, implying that an individual doesn't see well even with eyeglasses, contact focal points, medication, or medical procedures [Fig (1.12)].

Around the world, between 300 million and 400



Figure (1.12) "A visually impaired

A million individuals are outwardly weakened because of different causes. Around 50 million individuals in this gathering are visually impaired, incapable of seeing the light in one or the other eye. The vast majority of blindness happens in individuals more than 50 years of age.

Symptoms of Blindness in infants

A kid's visual framework starts to create in the belly. It doesn't wholly frame until around two years old. By 6 to about two months old enough, he ought to have the option to fix their look on an object and follow its development. His eyes should be appropriately adjusted by four months old enough and not turned internally or outward.

Signs and symptoms:

All individuals who are visually impaired or have visual hindrances have the primary indication of trouble seeing. Individuals with comparable degrees of visual misfortune may react differently to that side effect. Suppose one is brought into the world visually impaired. In that case, there is considerably less acclimation to a non-seeing world than for individuals who lose their vision late throughout everyday life, where there might be a restricted capacity to adapt to that visual misfortune. Emotionally supportive networks accessible to people and their mental cosmetics will likewise adjust the side effects of the absence of sight. Individuals who lose their vision abruptly, instead of throughout the long term, additionally can have more trouble changing following their visual misfortune. Related manifestations, such as uneasiness in the eyes, attention to the eyes, unfamiliar body sensation, and agony in the eyes or release from the eyes, might be available or missing, contingent upon the hidden reason for blindness.

A visually impaired individual may have no noticeable indications of any irregularities when sitting in a seat and resting. Notwithstanding, when blindness results from the cornea's contamination (the vault before the eye), the typically clear cornea may get white or dim [Fig (1.13)], making it hard to see the hues. In blindness from the waterfall, the Ordinarily, dark students may seem white.



Figure (1.13) "A blind kid with dim cornea"

Contingent upon the level of blindness, the influenced individual will show visual misfortune signs when endeavoring to move around. Some visually impaired individuals have figured out how to take a gander at the individual they are talking with, so it isn't obvious they are visually impaired.

Causes of Blindness

The numerous reasons for Blindness contrast with the country's financial state being considered. As shown in [Fig (1.14)], the primary sources of blindness in the created nations include:

- Cataract
- Ocular complications of diabetes,
- Macular degeneration,
- Glaucoma
- Traumatic injuries.
- Corneal scar
- Ancho.
- AMD

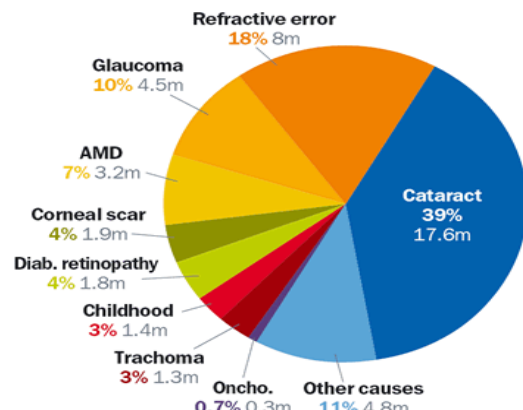


Figure (1.14) "Causes of blindness and how much they exist"

In third-world nations where 90% of the world's

Visually impaired population lives. The principal causes are infections, cataracts, glaucoma, injury, and inability to obtain glasses. In developed nations, blindness is not used to describe those people whose vision is correctable with glasses.

Infectious causes in undeveloped areas of the world cover:

- trachoma
- onchocerciasis (river blindness).
- Leprosy

The most well-known irresistible reason for blindness in created countries like **Egypt has herpes simplex**. Different reasons for blindness include:

- Nutrient and inadequacy.
- Retinopathy of rashness.
- Vein sicknesses include the retina or optic nerve, Stroke, and irresistible illnesses of the cornea or retina.
- Visual fiery illness.
- Retinitis pigmentosa.
- Essential or auxiliary malignancies of the eye, inherent variations from the norm, genetic sicknesses of the eye, and compound harming from poisonous specialists, for example, methanol.

If the problem is solved:

Egypt's society will be in a perfect state, and disabled people won't suffer from everyday challenges. Public health will be in good condition as there will not be any pollution. Blind people will find many places to work, and the economy of Egypt will increase. People with a disability perform as well as other employees, boosting staff morale and loyalty by creating a diverse workforce in Egypt.

If the problem is not solved:

The Blind will still be a burden on their communities since they are a non-productive class. Ignoring the Blind will affect the Egyptian economy adversely, as they are not a workforce. Blind will continue to be hurt by their sense of extreme incompetence, have not moved freely, tax their families, and will not be able to face a difficult life. If this problem is not solved, Arab progress will be delayed forever.

Topics related to the problem

Age-related Macular Degeneration (AMD):

Age-related Macular Degeneration is one of the primary sources of visual deficiency among seniors, influencing around 2.1 million individuals across the country. By 2050, it is assumed that the number will dramatically increase to 5.4 million. Individuals might be putting themselves in pointless danger of vision misfortune by failing to have sight-Saving eye tests. All of the Foundation of Ophthalmology's branches worldwide accentuate AMD mindfulness. It empowers individuals who are most in danger to guarantee their eyes' soundness by getting an eye test from an ophthalmologist – a doctor who represents considerable authority in the clinical and careful treatment of Eye infections and conditions.



Figure (1.15) "The difference between people with normal sight and people with AMD"

AMD is a degenerative sickness that harms the retina, the light-touchy tissue at the rear of the eye that centers pictures and transfers data to the cerebrum. Over the long run, retinal harm can prompt lasting loss of focal vision, restricting the capacity to drive, peruse and perceive faces [Fig (1.15)].

There are two structures, AMD – wet and dry [Fig (1.16)]. While the dry type of AMD prompts steady vision misfortune, the damp system advances faster. It is liable for 90% of all AMD-related visual impairment. Ongoing progressions in treatment choices have altogether diminished the occurrence of visual deficiency. Be that as it may, getting analyzed and starting treatment is essential. Quickly to ensure vision.

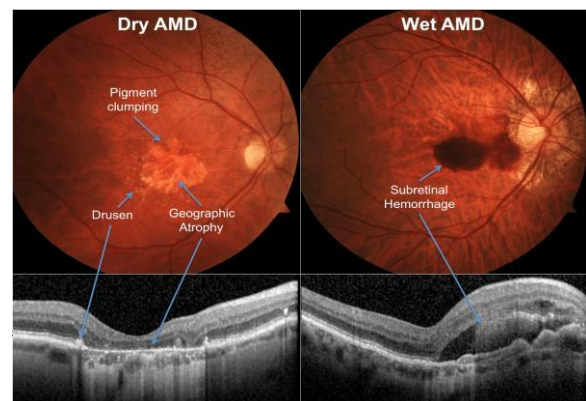


Figure (1.16) "Two types of AMD"

Air Pollution:

Openness to dirtied air is related to an expanded danger of glaucoma. This weakening eye condition can cause visual deficiency, as per an investigation. The discoveries, distributed in the diary *Insightful Ophthalmology and Visual Science*, show that individuals in neighborhoods with higher measures of fine particulate matter contamination were, in any event, six percent bound to report having glaucoma than those at all dirtied territories.

Air contamination indeed causes glaucoma [Fig (1.17)]. Glaucoma, a neurodegenerative illness, is the leading worldwide reason for irreversible visual deficiency and influences more than 60 million individuals worldwide. It most generally results from the development of pressing factors from the eye's liquid, harming the optic nerve that interfaces the look to the Mind

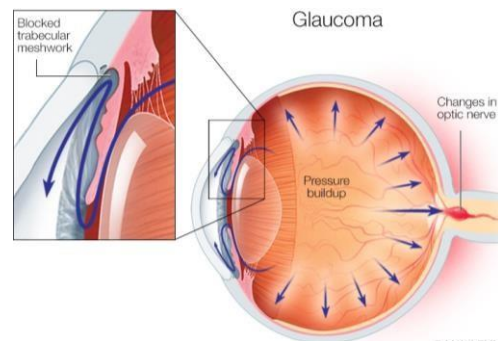


Figure (1.17) "Glaucoma"

Analysts put dazed individuals under a test to quantify intraocular pressure and A laser sweep of the retina to gauge the thickness of the eye's macula, the retina's focal territory. Along these lines, they established that individuals in the most- contaminated 25 percent of regions were, at any rate, six percent more bound to report having glaucoma than those at all dirtied quartile.

Eye pressure was not related to air contamination, which the scientists said recommends that air contamination may influence glaucoma hazard through an alternate instrument.

Air contamination might be adding to glaucoma because of veins' choking, which integrates with air contamination and connects to an expanded danger of heart issues. Another chance is that particulates may directly impact the sensory system and add to the aggravation.

Medical Mistakes:

As a whole, we trust doctors with our well-being. However, various blunders at any phase of a clinical cycle can prompt visual deficiency. For example, a few issues, such as illness, injury, and retinal separations, are distinguishable during routine tests. In any case, a specialist's inability to analyze a potential intricacy can prompt complete visual deficiency.

Tragically, babies can experience the ill effects of visual deficiency when specialists neglect to find a way to forestall genuine inconveniences upon entering the world. Visible lack brought about by ROP could be disposed of if clinical experts consistently screened for and treated it conveniently.

The expenses of adjusting to visual deficiency can be enormous. For instance, the individual may require extra clinical help. This may even incorporate a chaperone to focus on the individual 24 hours daily, seven days a week. That cost alone can run into large numbers. They may need to master new abilities to adjust and even change professional ways or quit working out and out. These anxieties have an incredibly passionate cost.

Topics Related to the Solution

Ultrasonic Sensors:

Ultrasonics are waves of greater frequencies than those of which humans can sense. Therefore, ultrasonic waves have frequencies greater than 20kHz.

Ultrasound waves:

The term "ultrasonic" is applied to sounds with frequencies above the audible range for humans. In other words, ultrasonic sounds include anything over 20kHz. For example, frequencies used for medical diagnostic ultrasound scans extend to 10MHz (megahertz) and beyond. However, many animals can hear sounds in the human ultrasonic frequency range. Furthermore, some animals, such as bats and dolphins

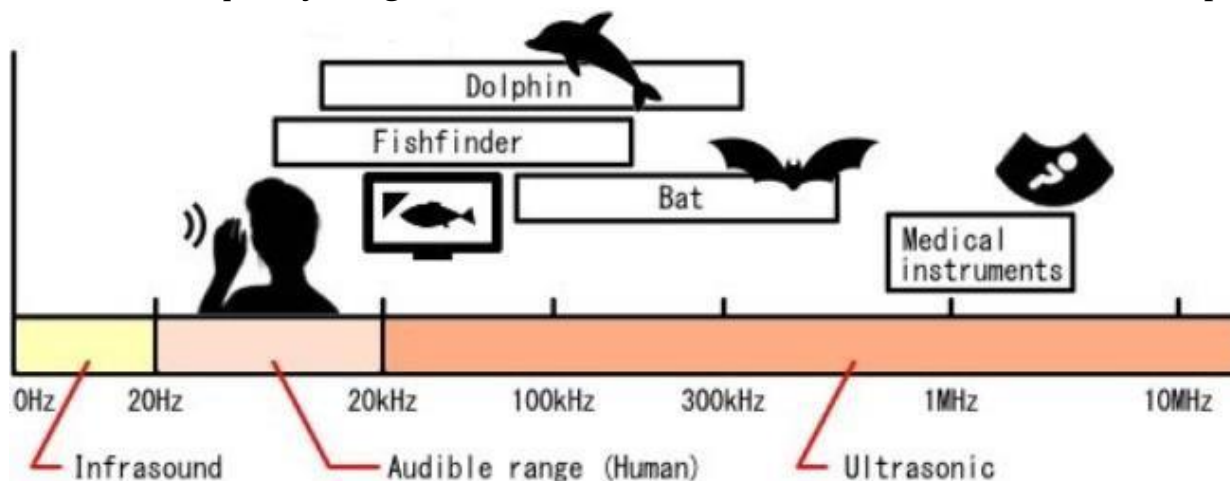


Figure (1.18) "sound frequencies and their applications"

depend on ultrasonic sounds to live. They do so by emitting ultrasounds and receiving their echo. By measuring the angle of the echo and the time interval between the transmitted wave and the received one, bats can find the target's distance. Likewise, dolphins use ultrasounds to understand the surrounding environment and communicate with other dolphins. [Fig (1.18)] shows different sound frequency ranges and some of their applications.

Properties of ultrasonic waves:

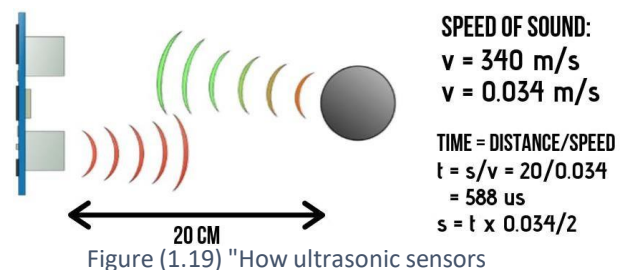
Ultrasonic waves have many properties that make them unique and usable in various applications. For example, ultrasonic waves have small wavelengths due to their significant frequencies. As a result, they can efficiently penetrate through various media.

Including gases, liquids and solids, but cannot travel through a vacuum. Also, ultrasonic waves travel at the speed of sound, which means their speed increases in denser mediums. Finally, ultrasonic waves have high energy content, which means they can be transmitted over a significant distance without losing much energy.

Ultrasonic sensors and how they work:

Ultrasonic sensing is one of the most reliable approaches through which humans can measure proximity and detect levels with high accuracy. An ultrasonic sensor is a device by which we can measure the distance to an object using ultrasonic sound waves. Ultrasonic sensors use transducers to send and receive ultrasonic pulses that are then used to gather information about an object's proximity, as shown in [Fig (1.19)].

In general, an ultrasonic sensor transmits an ultrasonic wave and receives its echo. It uses a transducer as a microphone to receive and send sound waves. After that, the sensor can determine the distance to the target by measuring the time-lapse between the sent and the received



Ultrasonic pulses. In summary, an ultrasonic sensor follows the rule in [Fig (1.22)] when measuring the distance to a specific target.

Ultrasonic sensors applications:

The ability to transmit and receive waves has given ultrasonic sensors plenty of substantial applications, some of which are discussed in detail in the following paragraphs.

Ultrasonic sensors in self-driving cars:

Many major auto manufacturers and technology companies are testing entirely sophisticated, self-driving vehicles. Nissan and GM say they will have completely self-driving cars on the road in the next few years. However, some companies, such as Tesla, have already started offering such cars since 2015. All self-driving cars use sensors extensively to monitor road and surrounding conditions. For example, ultrasonic sensors can detect cars within a specific range and then make decisions using AI technologies to decide the most reasonable route.

Tide gauge:

A tide gauge is a device that monitors sea level to detect high tides, storms, tsunamis and other coastal conditions. Ultrasonic sensors are used in tide gauges to detect real-time water levels, and, provided with an online database, the system can trigger an alarm in a risky situation.

Other Solutions Already Tried

We walk Smart Cane.

The walk smart cane is a white cane that uses innovative technology to help visually impaired people navigate their surroundings [Fig (1.20)]. The device has built-in speakers, smartphone integration, and sensors that send vibrations to warn users of obstacles up ahead. The walking cane
Comprises an electronic handle with a regular white cane inserted into the bottom. The electronic handle has an ultrasonic sensor to detect obstacles above the chest level and warn the user by vibrating the handle.

Visually impaired people can also pair the cane with their smartphone to integrate it with a voice assistant and Google Maps [Fig (1.21)]. It uses its built-in speakers to inform the user of nearby stores and infrastructural details that they cannot see.



Figure (1.20) "The weWALK smart cane"



Figure (1.21) "The weWALK cane Bluetooth connectivity"

Kursat Ceylan, the Turkish designer who is also blind, invented the cane to use modern technology as a tool for the visually impaired. Ceylan is the CEO and co-founder of the Young Guru Academy (YGA), the Turkish non-profit behind the invention. The technology used to manufacture the weWALK cane is built around an open platform, meaning third parties could add their smartphone-based functions. Its developers hope to eventually pair it with ridesharing apps and transportation services to improve its navigational abilities further.

" these days, we are talking about flying cars, but these people have been using just a plain stick, Kursat Ceylan explained to CNN. 'As a blind person, when I am at the metro station, I don't know which is my exit...I don't know which bus is

Approaching...which stores are around me. That kind of information can be provided with the walk."

The weWALK smart cane is also compatible with Android and iOS-based mobile phones [Fig (1.22)]. It features a USB input that can charge the battery with one full charge, lasting up to five hours of usage time. The cane is available on the weWALK original website and many other online stores for around 500 USD.



Figure (1.22) "The weWALK cane compatibly"

Points of strength:

- Detecting obstacles by an ultrasonic sensor and informing the user by vibration through the handle.
- Contains a compass, accelerometer, and gyroscope for integrating GPS technology and Google Maps.
- Bluetooth connectivity, to be paired with an iOS or Android smartphone and integration with apps such as Google Maps.
- Touch controls allow users to access their smartphone via the device without removing the phone from a pocket, purse, or backpack.
- Contains a microphone and speaker for voice control and Amazon Alexa integration.

Points of weakness:

- Does not attach to all types of canes that blind people use.
- Supports only Turkish and English languages.
- Battery life equals 5 hours only.
- Considered a bit expensive.

OrCam MyEye

For blind or visually impaired people, OrCam MyEye is an innovative voice-activated device that connects to practically any glasses [Fig (1.23)]. It can automatically read text from a book, smartphone screen, or any other surface. Furthermore, identifying faces,



Figure (1.23) "The OrCam smart device"

Helping visually impaired people shop independently, work more effectively, and live more independently [Fig (1.24)]. OrCam MyEye transmits visual information audibly in real-time and without the need for the internet.

The goal of OrCam is to leverage the power of artificial vision by integrating pioneering technology into a wearable device that enhances the lives of people who are blind, visually impaired, and have difficulty reading.

Prof. Amnon Shashua and Mr. Ziv Aviram, who are also the co-founders of Mobileye, the pioneer of the collision avoidance system and



Figure (1.24) "OrCam's facial recognition"

Autonomous driving innovator, jointly founded OrCam in 2010. In 2015, the first OrCam MyEye system was introduced, and in 2017, the next OrCam MyEye 2.0 generation was launched.

The mechanism of the innovative device works by an advanced optical sensor that captures an image of the environment, analyzes and identifies objects, faces, and reading texts, and then communicates the information audibly through a tiny speaker that rests above the user's ear or through a connected Bluetooth device. With the integrated voice assistance, the function of reading



Figure (1.25) "OrCam's reading ability"

The text could be activated by simply asking the device to "read the headlines, first paragraph, the article about, etc." or pointing to the desired part of the Text [Fig

(1.25)]. Also, the user can know what objects are in his surroundings by asking, "What is in the room? Or what's in front of me?".

Points of strength:

- Visually impaired people can now see with clarity, making their lives easier.
- small size, lightweight, can be attached to any glasses
- Ability to read text on any kind of material.
- Practical AI usage contains a face identification feature in real-time.
- With high accessibility, it responds to simple hand gestures.
- It can identify products as it comes with a database of millions of product bar codes and stores up to 150 users' desired products.
- Enough memory to include enough people in there.

Points of weakness:

- No audio recording to listen to the read text again at the moment.
- Extremely expensive, it costs about 4000USD.
- Battery life equals 2 hours only.

MYCIN

MYCIN is mainly a computer-based consultation system designed to assist physicians in the diagnosis of and therapy selection for patients with bacterial infections. In addition to the consultation system, MYCIN contains an explanation system that can answer simple English questions to justify its advice or educate the user. The system's knowledge is encoded in some 350 production rules that embody the clinical decision criteria of infectious disease experts. Much of MYCIN's power derives from these decision rules' modular, highly stylized nature, enabling the system to dissect its reasoning and allowing easy modification of the knowledge base.

The MYCIN system comprises three significant subprograms, as illustrated in [Fig (1.26)]. The Consultation Program is the system's core, which interacts with the physician to obtain information about the patient, producing diagnosis and therapy recommendations. The Explanation Program provides explanations and justifications for the

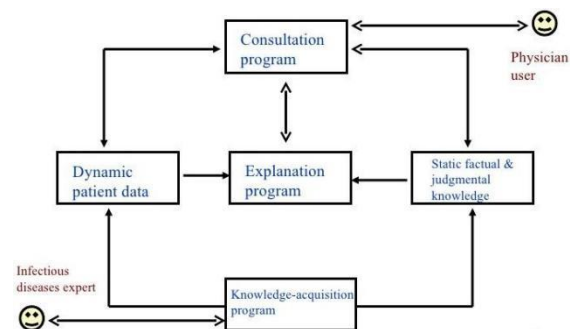


Figure (1.26) "The MYCIN Architecture"

Program's actions. The Knowledge-Acquisition Program is used by experts to update the system's knowledge base.

The Consultation Program:

Excerpts from a sample consultation are shown in [Fig (1.27)]. Following the double-asterisk prompt, the doctor's responses are all in the upper case. Much effort has gone into the system's human engineering aspects to make the interaction as smooth as possible. If the user finds a question unclear, he can request a rephrasing with a list of sample responses. An extensive dictionary of synonyms gives the user freedom

```

Marcuss-MacBook:src Marcuss$ ruby startmycin.rb
----- patient-1 -----
Patient's name: MARCUS BLOICE
Sex: MALE
Age: 39
----- culture-1 -----
From what site was specimen CULTURE-1 taken? ?
Must be one of: blood

From what site was specimen CULTURE-1 taken? BLOOD
How many days ago was this culture (CULTURE-1) obtained? ?
Must be a number

How many days ago was this culture (CULTURE-1) obtained? 3
----- organism-1 -----
Enter the identity (genus) of ORGANISM-1? WHY
[Why is the value of identity being asked for?]
Identity is one of the goal parameters.
Enter the identity (genus) of ORGANISM-1? ?
Must be one of: pseudomonas, klebsiella, entero, staphylo, bacteroides, strepto
Enter the identity (genus) of ORGANISM-1? ENTERO
  
```

Fig (1.27) "Excerpts from a MYCIN consultation"

In the choice of responses, simple typing or spelling errors are automatically corrected.

Production Rules:

MYCIN reasons about its domain using judgmental knowledge encoded as production Rules. Each rule has a premise, a conjunction of predicates over triples in the database. If the assumption is correct, the conclusion in the action part of The rule is drawn. If the premise is known

RULE035

PREMISE: (\$AND (SAME CNTXT GRAM GRAMNEG)
(SAME CNTXT MORPH ROD)
(SAME CNTXT AIR ANAEROBIC))
ACTION: (CONCLUDE CNTXT IDENTITY BACTEROIDES TALLY .6)

If: 1) The gram stain of the organism is gramneg, and
2) The morphology of the organism is rod, and
3) The aerobicity of the organism is anaerobic
Then: There is suggestive evidence (.6) that the identity of the organism is bacteroides

Fig (1.28) "MYCIN rule"

With less than certainty, the strength of the conclusion is modified accordingly. A typical rule is shown in [Fig (1.28)].

The control structure is a goal-directed backward-chaining of rules. At any given time, MYCIN works toward establishing the value of some clinical parameters. To this end, the system retrieves the (pre-computed) list of rules whose conclusions bear on this goal. The rule in [Fig (1.28)], for example, would be retrieved to establish an organism's identity. If, in evaluating the premise of one of these rules, some other piece of information that is not yet known is needed, MYCIN sets up a subgoal to find out that information; this, in turn, causes different rules to be tried. Questions are asked during the consultation when rules fail to deduce the necessary information. The rule is simply ignored if the user cannot supply the requested information. This control structure results in a highly focused search through the rule base.

Points of strength:

- One expert system may contain knowledge from more than one human expert, thus making the solutions more efficient.
- It's capable of explaining how it reached a solution.
- It helps to share human expertise.
- It's cheaper than consulting an expert for various domains, such as medical diagnosis.
- Fast response.

- Low error rate.
- Not affected by emotions, unlike humans.

Points of weakness:

- The significant problem of the expert system is common sense.
- It is developed for a specific domain.
- It needs to be updated manually. It does not learn itself.
- It isn't capable of explaining the logic behind the decision.
- It doesn't have human-like decision-making power.
- It can't possess human capabilities.
- It can't produce correct results from less amount of information.
- It requires excessive training.

Operations feedback systems

OFS is a software that aims to solve the biggest problem in manufacturing: wastage. Wastage means any time a business is getting less for more, whether throwing out product, making a poor quality product, not making enough product, filling out all the paperwork associated with all the time you are making less for more, and all the time wasted in analyzing inaccurate data



Fig (1.29) "OFS software"

To make poor decisions. The definition also illustrates how lost time could be invested while achieving the same results and profits through information feedback. That could be applied by dealing with OFS technologies or their set of products, as is apparent in [Fig (1.29)]

The best way OFS found that it will overcome the wastage problem is to provide manufacturers with accurate and valuable data through a set of products that automate data collection. This data can help manufacturers make intelligent decisions about OPEX and CAPEX, and understand the performance and output of human capital and machinery. OFS has successful case studies that indicate the critical role of feedback by information. They are presented in NCIA, Nudie Foods, Aryzta, PGH, and many other successful stories.

Points of strength:

- Reduce error-prone
- Save consuming manual data collection time
- Easily allows the observation of required data and insights
- Allowing businesses to remain viable
- It eliminates the need to use paperwork

Points of weakness:

- Its recommended solutions might not apply to specific businesses
- Not easy to use for everyone.

II. Generating and Defending a Solution

Solution Requirements

Cost:

The cost of the project is an important feature to be considered. The materials of the prototype should be as cheap as possible to be accessible. Also, while decreasing costs, the effectiveness of the AI recognition system must not be affected excessively. Therefore, the best result to accomplish is having the highest possible efficiency with the lowest possible cost.

Eco-friendliness:

The prototype can do its functions without harming the environment. In general, developed countries try to make a breakthrough in artificial intelligence to prevent as much environmental harm as possible. This is because AI-based projects hardly emit any harmful gases or by-products. So, this fact has to be maintained and achieved through the construction phase.

Availability:

It is how available the prototype is. This factor is crucial for the survival of any project, especially if it is introducing a new concept such as AI. This is achieved since the materials used are accessible and widespread. We also made sure that the materials do exist in Egypt in particular.

Practicality:

The prototype can be applicable in real life and in different environments or conditions. The prototype must have a high usage range. This can be accomplished since the used materials are widely available and made of high quality.

Usability:

It is how easily the prototype can be used. For a project to help blind people, wearing and using the device must be as simple as possible. This factor increases the project's popularity, thus benefiting both the user and the manufacturer. This condition can be achieved by designing an easy interface with clear buttons and tasks so users can use every feature.

Design Requirements

The Two main design requirements we focused on are:

Accuracy:

A highly efficient project guarantees higher probabilities of solving at least one of the aspects related to our specific problem. In our solution, efficiency expresses the accuracy of our prototype. In other words, if the prototype can recognize most of the surrounding symbols in different environments or conditions, high accuracy is achieved, and we can confirm that our project meets the desired design requirement. We aim to achieve more than 85% accuracy in object detection and recognition.

Time-Responding:

As the prototype will detect and recognize objects, the prototype should respond in an infinitesimal period. The estimated goal for the prototype is that it should work effectively in a real-life situation, which means that it should have a high prediction for any possible events and a fast response time technique. Our desire is to have less time to respond to this process. We aim to have a time response range from 3 to 5 seconds.

Selection of solution

In selecting a solution, the main aim is to produce an artificial intelligence-based assistive technology to help blind people. This system provides them with optical character recognition (OCR), object detection, and recognition. However, the information derived from OCR and object detection is converted from printed text into spoken language. These features give the blind the ability to interact and navigate with their communities. Object detection alerts blind people to physical objects. It helps them interact with their environment safely by providing data about the obstacles to avoid them. Also, OCR plays a crucial role in converting handwriting and printed text to a machine-learnable format.

About 4.25% (285 million) of the world's population experience visual issues. Furthermore, 14% of them (39 million) are entirely blind. That leaves us with a serious problem to deal with. Blind people are considered non-productive. Thus, our economy can be significantly impacted by ignoring them. Every day, blind people face many problems, becoming a burden on their friends and relatives. Going for a stroll should no longer be such a difficult job for someone else. In brief, Blind people deserve to have an everyday life.

Currently, visually impaired people use a traditional cane to direct them when they move from one place to another. Although the traditional cane is the most widespread means used today by visually impaired people, it cannot help them detect dangers from all levels of obstacles. We propose a new intelligent system for guiding individuals who are visually impaired or partially sighted in this context. The system enables visually impaired people to move quickly and confidently as sighted people.

The chosen prototype will be smart glasses, which do three main functions: text reading, object detection, and object recognition. The glasses' mechanism will be a process of three steps. The process's input is a digital image captured by the optical sensor, which is the camera of the user's smartphone. The mobile phone is connected directly to a Raspberry Pi via Bluetooth. The Raspberry Pi starts to process and analyze the image, state what object or text is in the image, and then auditory outputs are provided to the user through a headset connected to the Raspberry Pi.

Raspberry Pi:

The Raspberry Pi is a single-board computer (SBC). The board is a fully functional computer with a dedicated processor and memory [Fig (2.1)]. It can run an operating system (runs on Linux). Raspberry Pi includes its USB ports audio output and has a graphic driver for HDMI output, showing how it can run multiple programs. Raspberry Pi can do any task a traditional computer can do. The Raspberry Pi is



Figure (2.1) "the Raspberry pi

We considered the prototype's core as our code will be stored on it, allowing it to analyze the image and decide what object or text confronts the visually impaired person. Thus, the Raspberry Pi will present a solution with the highest possible accuracy at a low cost, which is our design requirement.

Ultrasonic Sensor:

It plays a crucial role since it acts as the eyes for visual impairments. Its primary job is to detect objects in the wearer's path by emitting sound waves and measuring the time it takes for these waves to bounce back after hitting an object. The sensor calculates the distance between the wearer and the object. This real-time distance data helps identify obstacles like chairs, tables and walls. When the sensor detects an object, it triggers a signal to the Raspberry Pi. The Raspberry Pi, the brain behind the operation, processes this information, interprets the signal and alerts the wearer through the connected headphones. To sum up, the ultrasonic sensor acts as the guiding light for the visually impaired, helping them navigate their surroundings safely and independently. As shown in Figure (2.2)



Figure (2.2) “the ultrasonic sensor

VR Goggles:

Equipped with the ultrasonic sensor, the Raspberry Pi 3. It is used as the integral body the patients wear to hold the components together for simple use, better attachment, and better appearance.

Headphones:

Output device for reading text and object recognition aloud to the user through sound.

III. Constructing and Testing a Prototype

Materials and Methods

ITEMS	DESCRIPTION & QUANTITY	PRICE	IMAGE
Raspberry Pi	It is a low-cost, credit-card-sized computer	2000 LE	
Breadboard	A thin plastic board holds electronic components that are wired together.	60 LE	
Jumper Wires	10 Male – Male 10 Female – Female 10 Female – Male	16.5 LE	
Ultrasonic Sensors	Two sensors send and receive ultrasonic data to notify the blind person if there are any obstacles	80 LE	
VR Goggles	Only the frame attaches all the components to it, so the blind person can wear it	100 LE	

Headphones	Used as the tool to receive the output in the form of sound. -	100 LE	
Battery	Source of electricity	70 LE	
Memory Card	16 gigabytes, considered as the hard drive for the microcomputer (Raspberry) to restore the data of our code	80 LE	
raspberry-pi-webcam	A webcam is a small camera for video and audio used in online communication and virtual meetings.	180 LE	

Table (3.1) "Materials; cost and description"

Methods:

1. The code was constructed with Python programming language on the visual studio, developed many times and tested to see if there were any errors, as shown in [Fig (3.1)]
2. The Raspberry OS was downloaded and burned to the memory card through any computer.
3. The memory card was put into the Raspberry Pi, which was later connected to a power bank, a monitor, a keyboard and a mouse.

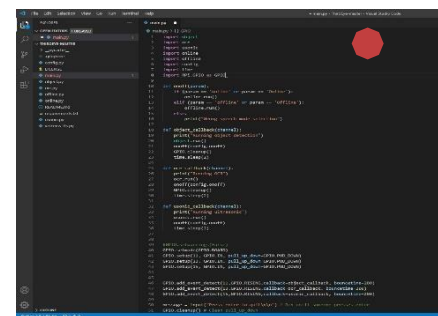


Figure (3.1) the main code

4. The Raspberry was turned on and set up to be ready to read Python, and then the code was downloaded.

5. The two ultrasonic sensors and the buttons were connected to the breadboard by jumper wires. The breadboard was connected to the Raspberry by jumper wires, as shown in [Fig (3.2)].
6. The components were later fixed to the VR goggles to make it easier for the visually impaired person to walk freely with the prototype.
7. A mobile phone was connected to the Raspberry via Bluetooth to be used as an optical sensor for object and text detection.

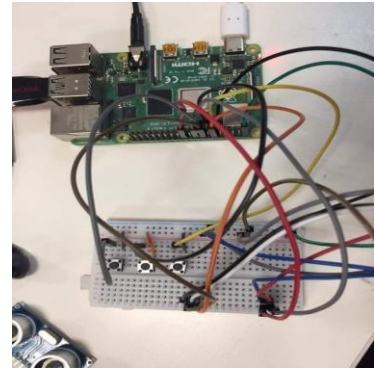


Figure (3.2) the 5th step

8. The headset was connected to the Raspberry through the audio jack so the visually impaired person could hear the alerts of the near obstacles and the audio of any written text around him.
9. The prototype was later trained on models of everyday objects and written texts to identify anything that might confront the visually impaired person.
10. Finally, the prototype was tested again to see if everything worked correctly and the connections were valid.
11. A stopwatch was used to calculate and estimate the OCR and object recognition response time.

Safety Precautions:

The prototype construction process was entirely within a safe environment as we were committed to the safety precautions:

1- User Well-Being:

Ensure the innovative glass design prioritizes user comfort, avoiding sharp edges, uncomfortable pressure points, or materials that may cause skin irritation.

2- Emergency Stop Mechanism:

Implement an emergency stop or manual override feature that allows users or testers to quickly halt any autonomous actions if they feel uncomfortable or unsafe.

3- Battery Safety:

Ensure the intelligent glass's battery and charging mechanisms are safe and comply with industry standards to prevent overheating, explosions, or other risks.

Test plan

After we had finished our prototype, we started doing our test plan to make sure of our idea and to ensure we would achieve our desire. So, some steps were made for the test:

- 1) We will create scenarios simulating various environments (indoor, outdoor, crowded spaces) to evaluate the intelligent glass's ability to detect and recognize objects.
- 2) We will Test the smart glass's text recognition capabilities with different fonts, sizes, and styles to ensure versatility.

The accuracy of the prototype was measured by exposing it to several real-life situations where it confronts unknown text passages/objects:

- 1) Analyzing how many words/objects it read correctly.
- 2) The accuracy percentage was calculated using the following equation:

$$\text{Accuracy} = \frac{\text{The no. of Detected words or objects}}{\text{The total no.}} \times 100$$

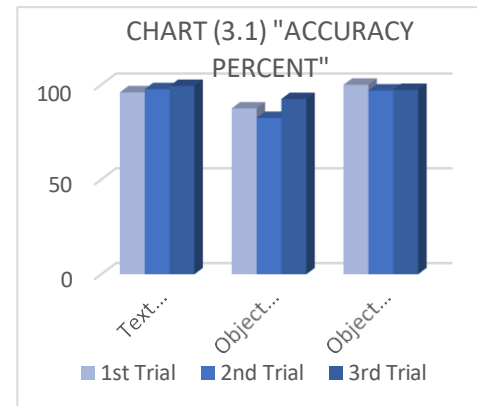
Data Collection

Results were recorded after testing the prototype many times, and both negative and positive results appeared.

Negative results:

The first time constructing the prototype, the results were unsatisfying due to some construction issues, such as incorrect wiring connections. After those connections

We were modified as they were written in the methods section, and positive results appeared, as shown in the following graphs and calculations.



Positive results:

The prototype achieved an average accuracy percentage of 97.66% in reading text, 87.5% in detecting objects and 98.1% in recognizing them. Also, the prototype achieved a lower cost of 81.5% compared to its global competitor (weWALK smart cane).

TRIAL	CORRECTLY DETECTED WORDS	TOTAL WORDS	ACCURACY PERCENTAGE	Time Response(s)
1 ST TRIAL	288	300	96%	4.31s
2 ND TRIAL	293	300	97.6667%	3.92s
3 RD TRIAL	298	300	99.3334%	3.81s
Average	293	300	97.6%	4.01s

Table (3.2)" Text reading"

TRIAL	CORRECTLY DETECTED OBJECTS	TOTAL OBJECTS	DETECTION ACCURACY	CORRECTLY RECOGNIZED	RECOGNITION ACCURACY	Time response(s)
1 ST TRIAL	35	40	87.5%	35	100%	4.40 s
2 ND TRIAL	33	40	82.5%	32	96.97%	4.62s
3 RD TRIAL	37	40	92.5%	36	97.29%	4.48s
Average	35	40	87.5%	34	98%	4.5s

Table (3.3) "Object detection and recognition"

IV. Evaluation, Reflection, Recommendations

Analysis and Discussion

The World Health Organization (WHO) estimated that globally, approximately 45 million people are blind. In comparison, around 285 million people have moderate to severe visual impairment. About 1.3 million people in Egypt had moderate to severe vision impairment. Therefore, the project will show a great demand due to the high portion affected by this ordeal. Leading to operating the factories to keep up with improvements and the latest version of this primary model. In addition, facilities served to patients ensure equality, deal with various situations, and accomplish tasks. Preserving public health and urging industrialization. Moreover, patients could be a practical part of the community that could help in industry development.

Computer vision is a field of artificial intelligence that trains computers to interpret and understand the visual world. Using digital images from cameras and videos and deep learning models, machines can accurately identify and classify objects and then react to what they "see." Along the same lines, our smart glasses work perfectly. The process's input is a digital image captured by the optical sensor, which is the camera of the user's smartphone. The mobile phone is connected directly to the Raspberry Pi via Bluetooth (PH.3.04&PH.3.05) *. The Raspberry Pi starts to process and analyze the image, state what object or text is in the image, and then provide auditory outputs to the user through a headset connected to the Raspberry Pi. We discuss artificial neural networks in the following paragraphs to better understand how all these components are connected.

Artificial Neural Networks: What, How and Why?

First: What Neural Networks Are:

As the name suggests, neural networks simulate what science knows about human brain function. It is a computing unit formed by a combination of small but highly interconnected elements called "neurons". Neurons are units that hold a specific value and are arranged in distinct layers. The first layer contains the input neurons, the pixels of an image.

for example. The last layer is a combination of all the possible outcomes, possible predictions of

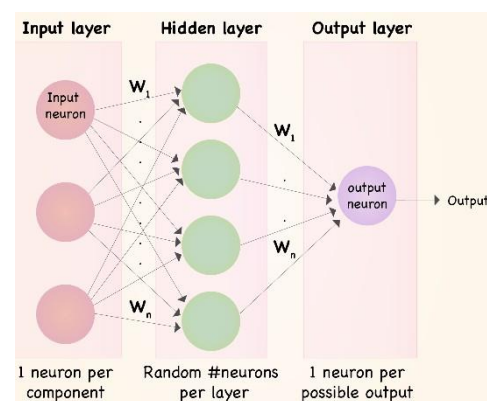


Figure (4.1) "A simple visualization of a neural network"

What the image can be. The layers in between are called "hidden layers," where all the processing happens. The number of hidden layers differs from one system to another according to the variety of inputs and their probabilities. Each layer is connected to the next layer through so-called "channels." Channels are characterized by weights and biases (commonly referred to as "W" and "b"). Weights are random real values given to each input neuron to express how vital the corresponding input is to the output. Meanwhile, biases are constants that act as intercepts in linear functions. They aim to give the model more accuracy and flexibility by providing a wider variety of variables to modify. The system adjusts weights and biases to reach the most perfect combination that suits the outputs. This structure, shown in [Fig (4.1)], introduces inputs to the computing unit in the first layer. After that, it goes through multiple processes before coming up as outputs [Fig (4.1)]. But how are our inputs changed to be output?

Second: How Neural Networks Work:

For inputs to be outputs, they must undergo some processing. When an input reaches the hidden layers, it passes through what is called an activation function (sigmoid function) [Fig (4.2)]. The sigmoid function ($\frac{1}{1+e^{-z}}$) tests whether an input is active or not. It takes

in $\sum(w \cdot x + b)$ and outputs a value between 1 and 0 (MA.3.01) *, where more negative values approach 0, thus considered inactive, whereas more positive values approach 1, thus considered active [Fig (4.2)]. Generally, the output of the activation function in one hidden layer is given as an input to the next neuron in the next hidden layer. This process called forward

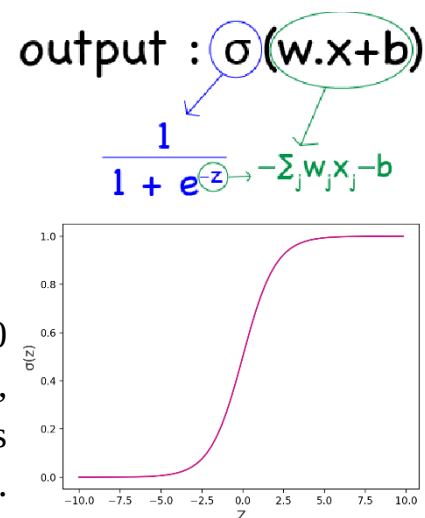


Figure (4.2) "Sigmoid function and graph"

propagation, continues to happen until the system reaches the output layer.

Third: Why Neural Networks are able to learn:

The gripping strength of machine learning algorithms is their capability to develop and give improved outputs every time. All machine learning models experience a training phase in which the system compares the real or desired output of the Training inputs with the predicted output. During this phase, the model adjusts the values of weights and

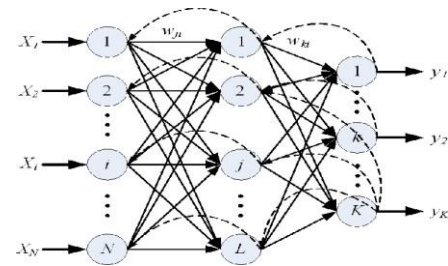


Figure (4.3) "Backward Propagation"

Biases go through a process called backward propagation. The error is calculated in this process, and the propagation starts from the output layer [Fig (4.3)]. While moving from one layer to another, weights and biases change until the input layer is reached. Then, the model tests the training inputs again. The process continues until the output from the network approximates the real value for all the training inputs.

Ultrasonic Sensors:

In addition to detecting texts and images, the smart glasses can alarm the user if any obstacles are on his road, and this would have never been achieved without attaching an ultrasonic Sensor. It is a device by which we can

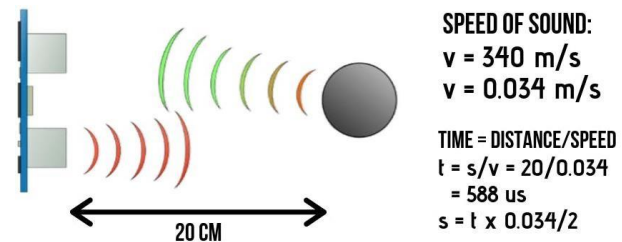


Figure (4.4) "How ultrasonic sensors work"

measure the distance to an object using ultrasonic sound waves (**PH.3.01&PH.3.09**)
 *. Generally, an ultrasonic sensor uses transducers to transmit an ultrasonic wave and receive its echo. After that, the sensor can determine the distance to the target by measuring the time lapse between the sent and the received ultrasonic pulse. In summary, an ultrasonic sensor follows the rule in [Fig (4.4)] when measuring the distance to a specific target.

Programming and Libraries:

First, OCR:

Python-tesseract is an optical character recognition (OCR) tool for Python. It will recognize and "read" the text embedded in images.

Python-tesseract is a wrapper for Google's Tesseract-OCR Engine. It is also helpful as a stand-alone invocation script to Tesseract, as it can read all image types supported by the Pillow and Leptonica imaging libraries.

Meanwhile, OCR accuracy depends on the text's size and distance.

Second, Object Detection:

The COCO dataset used: The COCO (Common Objects in Context) dataset classes are divided into two main categories: "things" and "stuff."

Also, the accuracy of detection depends on video resolution.

"Things" classes include objects easily picked up or handled, such as animals, vehicles, and household items. Examples of "things" classes in COCO are Person, Bicycle, Car, and Motorcycle. "Stuff" classes include background or environmental items such as sky, water, and road. It contains over **330,000** images, each annotated with **80** object categories.

Recommendations

GPS Navigation:

GPS work using triangulation interaction between three or more satellites, as shown in [Fig (4.5)]. Each GPS satellite sends a great sign and orbital boundaries that permit GPS gadgets to interpret and process the satellite's precise area. According to our smart glasses, GPS technology furnishes the disabled, visually impaired one with the easiest method of getting to any location he needs. This technology will be actualized by providing the glasses with an informational set of data from Google Maps about all shops, clubs, or

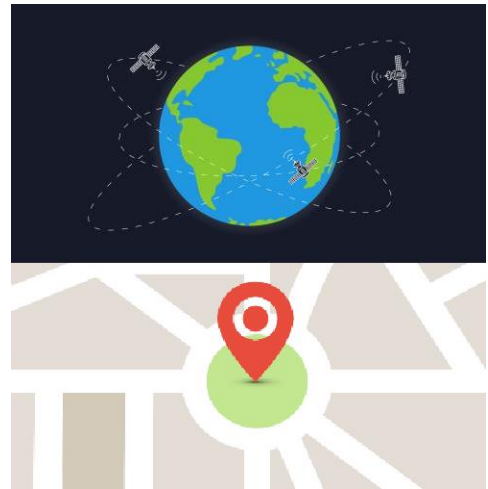


Figure (4.5) "Triangulation interaction"

restaurants close to where the patient lives. Besides, the patient can likewise effectively set up his current co-ordinates and offer to his contacts.

Voice Assistance:

The speech recognition system receives the voice and turns it into text by separating a discourse recording into singular sounds, examining each sound, utilizing logarithms to locate the most likely word fit in that language, and deciphering those sounds into text, [Fig (4.6)]. This text can be used as an input to the smart glasses. The text is put under



Figure (4.6) "Voice assistance"

a processing called natural language processing (NLP); via AI, the intelligent device does what the user ordered accurately, correctly, and faster than humans.

Phonak Roger Pen:

The Roger Pen is a portable microphone used in conjunction with the hearing aids, as shown in [Fig (4.7)]. It allows the user to hear and understand sounds both in noise places and distance speech. It instantly transmits the signals via high-quality sound and can connect to cell phones and other audio outputs for TV and multimedia listening through Bluetooth. This

device could be used along with speech recognition system for the ultimate voice assistance experience.



Figure (4.7) "Phonak Roger Pen"

Learning Outcomes

SUBJECT	LEARNING OUTCOMES	DESCRIPTION	HOW WE BENEFITED FROM IT
PHYSICS	PH.3.01	This LO discusses waves, their types and how they travel in different mediums.	It helped understand what type of waves ultrasonic sensors depend on.
PHYSICS	PH.3.04	This LO discusses the fundamentals of communication and important vocabulary	It helped us understand how we can establish a connection between our smart glasses and the user
PHYSICS	PH.3.05	This LO is a sequel to the previous one. It discusses the concept in further details adding some real-life applications	It helped us understand how the connection is established through Wi-Fi and Bluetooth.
PHYSICS	PH.3.08	This LO discusses the properties of waves of different types	We could understand the properties of ultrasonic sound waves in our ultrasonic sensors, and we could know more about how they work.
Biology	Bi.3.01	This LO discusses neuron structures and how it works in nervous system	It helped us understand how artificial neural network works
MATHS	MA.3.01	This LO discusses functions, their derivatives and properties.	After studying this LO, we could know more about what sigmoid functions are and how they work.
Statistics	ST.3.01	This LO discuss probability and how to make normal distribution graph	This LO helped us to know the probability of success or failure in certain situation

Statistics	ST.3.02	This LO discuss correlation and regression	This LO helped us in making our AI model and to set some databases having some predicted values with small percentage error.
ENGLISH	EN.2.05	This LO discusses academic writing skills and why they are important	After studying this LO, we knew how to write in an academic way and this helped us convey our ideas with more flexibility.
COMPUTER SCIENCE	CS.1.02	This LO discusses fundamentals of programming	After studying This LO, we were prepared to start our journey in programming.
COMPUTER SCIENCE	CS.1.03	This LO discusses Microsoft Word and how to use it to create,	After studying this LO, we were able to document all our

Table (4.1) "Learning outcomes"

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